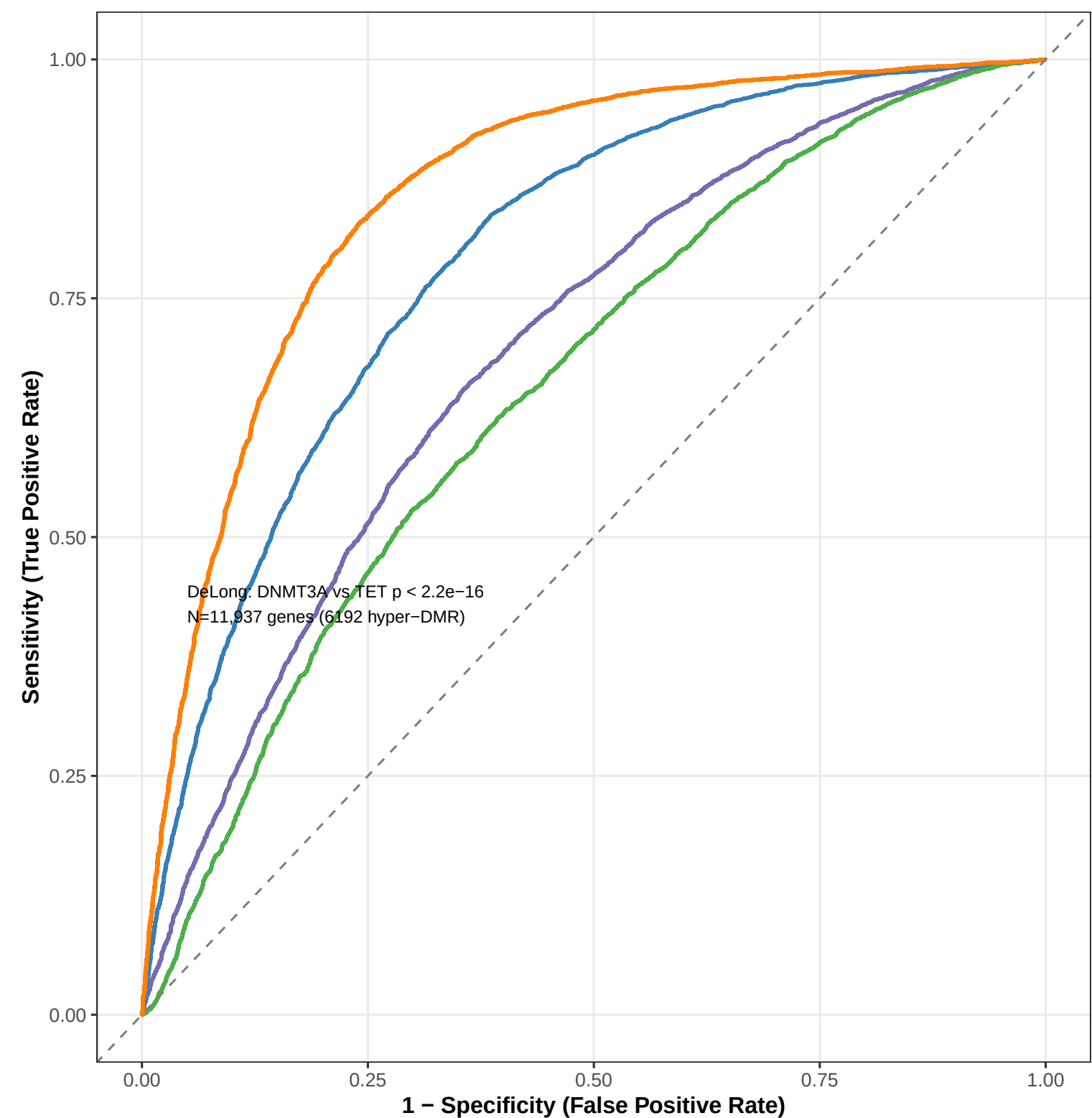


DNMT3A Recruitment vs TET Impediment: Model Comparison

N=11,937 genes | TET outperforms DNMT3A (DeLong p < 2.2e-16 shared, p < 2.2e-16 exclusive)

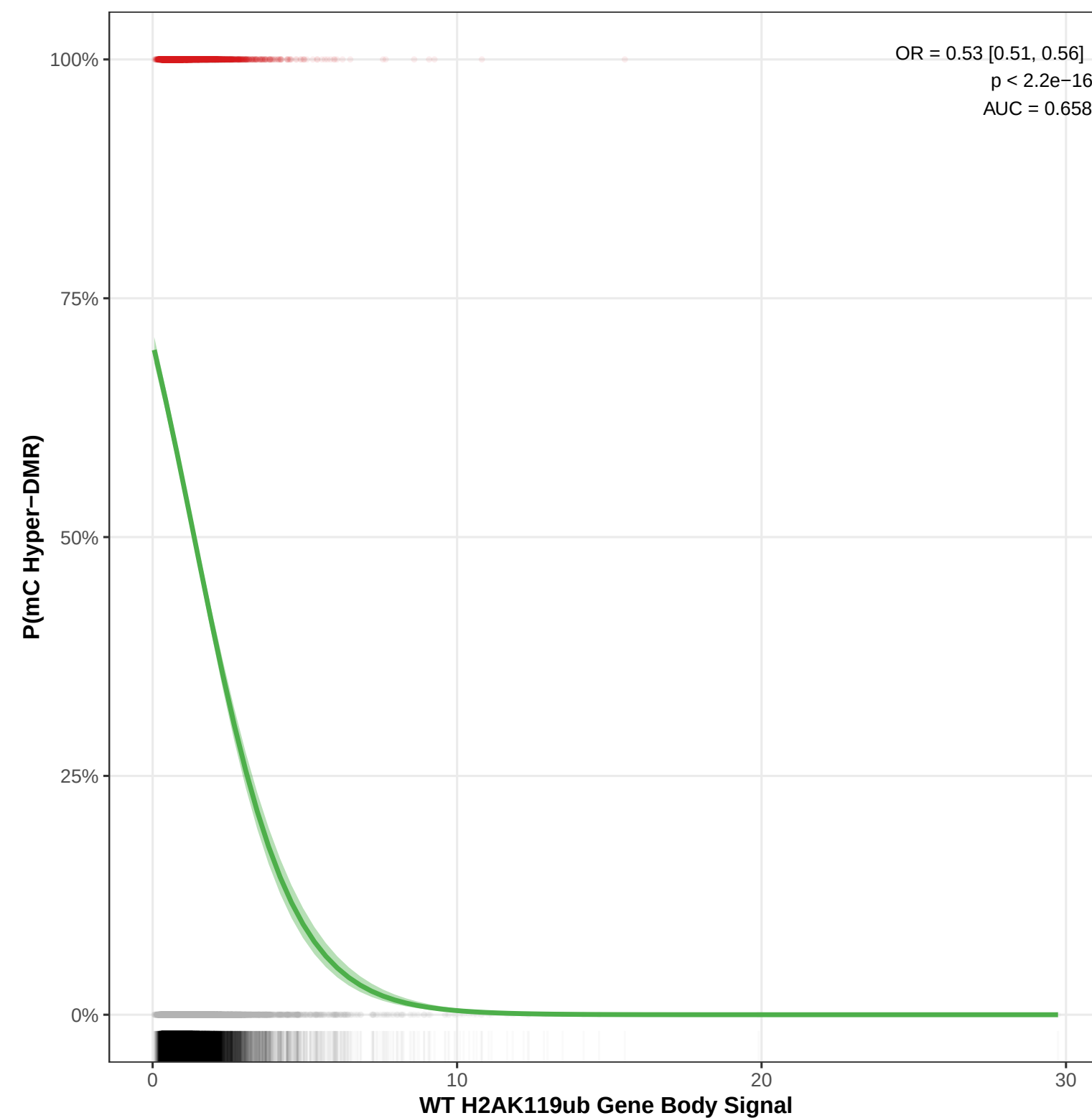
A. ROC: 5 Model Comparison



Full (AUC=0.861) DNMT3A recruitment (AUC=0.696) TET impediment (AUC=0.793) K119ub only (AUC=0.658) Stepwise (AUC=0.861)

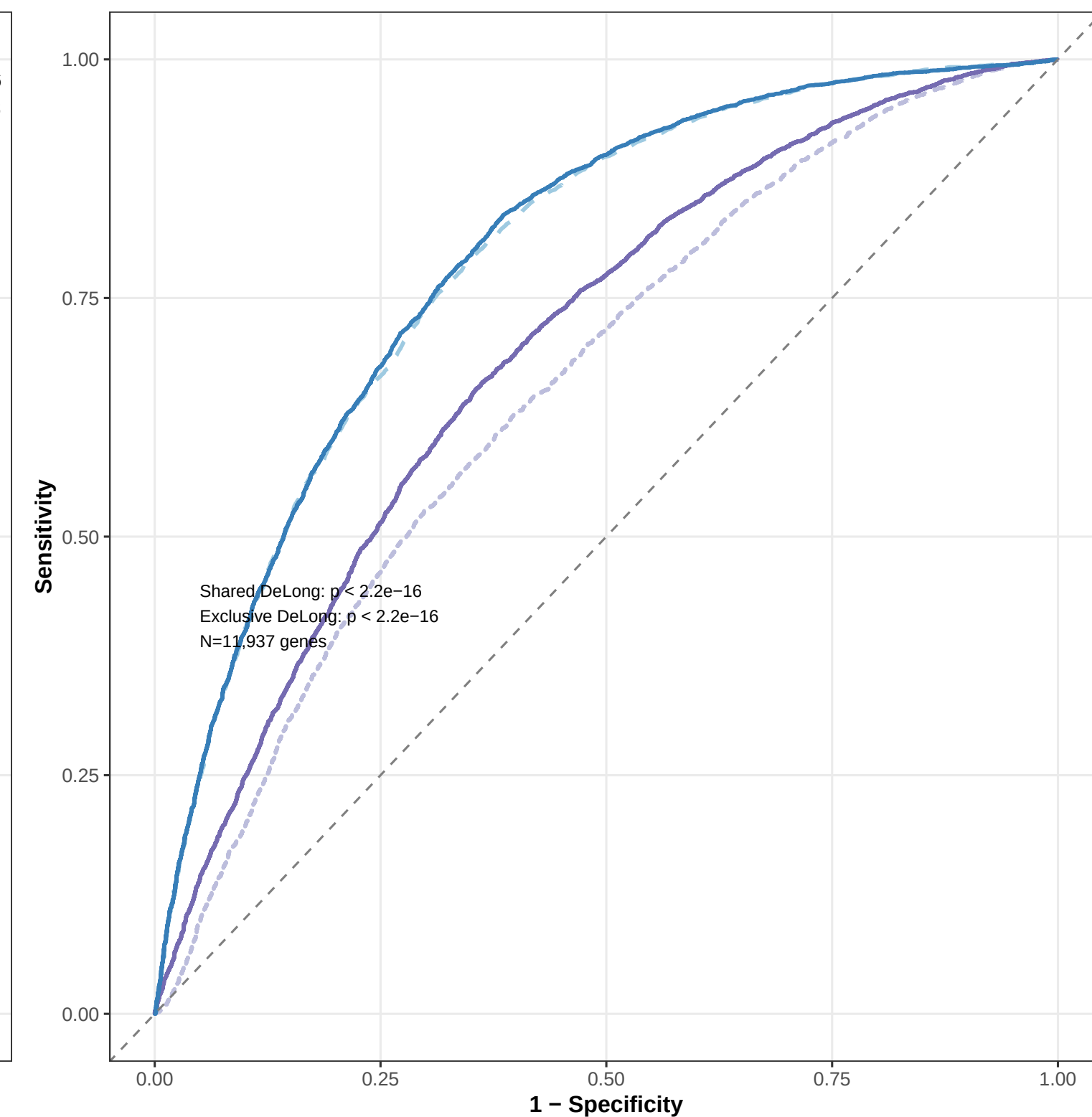
Full: K119ub + ATAC + CpG dens. + 5mC + 5hmC + length + expr | DNMT3A: K119ub + ATAC + CpG dens. | TET: 5hmC + ATAC | K119ub: K119ub only | Step: AIC-selected

B. K119ub Predicted P(hyper-DMR)



mC DMR Status Not hyper-DMR Hyper-DMR

C. Exclusive Model ROC



DNMT3A (shared) (AUC=0.696) TET (shared) (AUC=0.793) DNMT3A (exclusive) (AUC=0.658) TET (exclusive) (AUC=0.790)

Full: K119ub + ATAC + CpG + 5mC + 5hmC + length + expr | DNMT3A: K119ub + ATAC + CpG | TET: 5hmC + ATAC | DNMT3A excl: K119ub + CpG | TET excl: 5hmC only